Customer Behavior Analysis Using SQL & Python

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- Project Type: Data Analytics Project
- Technologies: Python, MySQL, Pandas, SQLAlchemy, TextBlob

Problem Statement

- low customer in-use website and marketing
- Buying customers are coming at a lower percentage
- I will spend an amount on digital marketing, but the revenue will not increase.
- Customer review is not proper, so that is not understood.
- Customers leaving negative reviews about certain products.
- Low repeat purchase rates, indicating dissatisfaction.
- A lack of structured insights into customer opinions.

Project Objectives

- 1. Analyze customer demographics and distribution
- 2. Evaluate product performance using ratings
- 3. Measure engagement across campaigns and content types
- 4. Perform sentiment analysis on customer reviews

Project Architecture

Pipeline:

CSV Data → MySQL Database → Create tables → Analysis → Sentiment analysis → Get a review sentiment output

Technologies Used

- Python – Data processing
- MySQL – Database storage
- Pandas – Data manipulation
- SQLAlchemy – Database connection
- TextBlob – Sentiment analysis
- SQL – Analytical queries

Questions :

1. What factors are influencing customer engagement?
2. At what stage are customers dropping off in their journey?
3. How do customer reviews impact purchasing behavior?
4. Which products, locations, and customer segments are performing well?
5. Analyze customer sentiment from the reviews.
6. identify key complaints and improvement areas.
7. Find patterns between negative reviews and product performance.
8. Recommend strategies to enhance customer satisfaction.

Dataset Overview

Datasets Used:

- customers.csv – customer information
- products.csv – product catalog
- customer_reviews.csv – product reviews
- engagement_data.csv – marketing engagement
- customer_journey.csv – interaction history
- geography.csv – location data

Dataset Summary

- Total Customers: 100
- Total Products: 20
- Total Reviews: 100
- Average Rating: 3.73

Data Loading Process

File: data_loading.py

Steps:

- 1. Connect to MySQL using SQLAlchemy
- 2. Load CSV datasets using Pandas
- 3. Insert data into MySQL tables using to_sql()
- 4. Prepare database for analytics

Database Design

Tables created in MySQL

File: Create_table.sql

- customers = who bought
- products = what they bought
- orders = The transaction (connects customer + product)
- Reviews = Feedback (connects customer + product)

- These tables help analyze customer behavior across multiple dimensions.

SQL Analytics Layer

file: analysis.sql

1) Total customers:

→ This query calculates the total number of registered customers in the system, which helps understand the overall user base.

2) Total products:

→ This query calculates the total number of products available on the platform.

3) Total reviews + avg rating:

→ This query evaluates overall product satisfaction by calculating the total number of reviews and the average rating

4) Top 5 products by avg rating:

→ This query identifies the best-performing products based on customer ratings.

5) Bottom 5 products by avg rating:

→ This query helps identify poorly performing products that may require quality improvements.

6) Overall engagement (proxy metric):

This query helps identify poorly performing products that may require quality improvements.

7) Engagement by Content Type:

This query compares engagement across different content types such as video, images, and blogs.

8) Top 10 Products by Engagement:

This query identifies the products generating the most customer engagement.

9) Top 10 Campaigns by Engagement:

This query helps marketing teams identify which campaigns drive the highest engagement.

10) Customers by geography (if customers has GeographyID):

This query analyzes the geographical distribution of customers across different regions

Automation Layer

- file: run_sql_analysis.py

➢ This script automatically:

➢ Reads SQL queries from analysis.sql

➢ Executes them in MySQL

➢ Prints analytical results in the terminal

Sentiment Analysis Layer

file: sentiment_analysis.py

Steps:

- Load review data from database
- Apply TextBlob sentiment scoring
- Classify reviews as Positive, Negative, or Neutral
- Export results to CSV file

Business Insights

- Identify most liked products
- Detect underperforming products
- Understand which marketing campaigns perform best
- Analyze customer satisfaction from review sentiment

Top Rated Products → Product 8 (5.00)
Lowest Rated Products → Product 7 (2.67)
Geography distribution →Geography ID 4 → 18 customers(highest)

Engagement Summary →

Total Engagement Records : 100
Total Likes : 10,446
Average Likes per Record : 104.46

Content Performance →

Video : 3,473 likes
Blog : 2,999 likes
Social Media : 2,594
likes Newsletter : 1,380 likes

Sentiment Analysis →

Positive Reviews : 84
Negative Reviews : 11
Neutral Reviews : 5

Project Workflow

- Step 1: Load datasets into database
- Step 2: Run SQL analytics queries
- Step 3: Automate query execution
- Step 4: Perform sentiment analysis
- Step 5: Generate insights

Conclusion

- Customer behavior was analyzed using SQL and Python.
- Product performance, engagement trends, and geographic insights were identified.
- Sentiment analysis provided deeper understanding of customer opinions.
- The insights help businesses improve marketing strategies, product quality, and customer experience.
- The project demonstrates how data analytics can support data-driven decision making.

Business Recommendations

- Allocate more marketing budget to video content because it has the highest engagement.
- Redesign newsletter campaigns since they receive the lowest likes.
- Promote high-performing products like Product 8 and Product 1
- Improve low-rated products, especially Product 7 and Product 16.
- Replicate the successful strategy used in Campaign 17
- Focus promotions in GeographyID 4, which has the largest customer base.
- Optimize the checkout process to reduce customer drop-off.
- Analyze negative review sentiment to fix product issues.